FML Assignment 4

Raju Rishikesh Gollapally

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#Summary

1. Range and Z-score scaling have been utilized to standardize the data.
2. When k-means clustering is done using the silhouette strategy, and their the best result is five clusters with five centroids.
3. In this case, the number we generated for the best k = 5 is around 0.3, indicating that it is good. The silhouette spans from -1 to +1, with higher numbers denoting greater quality.
4. The cluster summary and the data set's properties are used to classify the cluster names:

Cluster1 greater NetProfitMargin and "High Market Cap & High ROE"

"Hold CLuster" is Cluster 2.

"Low Market Cap & Less ROA" is Cluster 3.

"Small Net Profit Margin & High PE Ratio" is the fourth cluster.

"Moderate Buy & Moderate Sell" is Cluster 5.

#Problem Statement:

An equities analyst is studying the pharmaceutical industry and would like your help in exploring and understanding the financial data collected by her firm. Her main objective is to understand the structure of the pharmaceutical industry using some basic financial measures. Financial data gathered on 21 firms in the pharmaceutical industry are available in the file Pharmaceuticals.csv Download Pharmaceuticals.csv. For each firm, the following variables are recorded.

1.Market capitalization (in billions of dollars). 2.Beta. 3.Price/earnings ratio. 4.Return on equity. 5.Return on assets. 6.Asset turnover. 7.Leverage. 8.Estimated revenue growth. 9.Net profit margin. 10.Median recommendation (across major brokerages). 11.Location of firm's headquarters. 12.Stock exchange on which the firm is listed.

```
library(readr)
input_pharmaceuticals <- read.csv("Pharmaceuticals 2.csv")
View(input_pharmaceuticals)
```

#examining pharmaceutical data #installing the tidyverse library to manipulate data

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.3     v purrr     1.0.2
## v forcats   1.0.0     v stringr   1.5.0
## v ggplot2   3.4.3     v tibble    3.2.1
## v lubridate 1.9.2     v tidyr     1.3.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

#the factoextra library has been installed in order to extract and visualise the data.

```r
library(factoextra)
```

```
## Warning: package 'factoextra' was built under R version 4.3.2

## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

#ISOLAR library loading for statistics instruction

```r
library(ISLR)
```

#utilizing FlexClust to execute techniques to clustering.

```r
library(flexclust)
```

```
## Warning: package 'flexclust' was built under R version 4.3.2

## Loading required package: grid

## Loading required package: lattice

## Loading required package: modeltools

## Loading required package: stats4
```

#installing the training dataset for the library

```r
library(caret)
```

```
##
## Attaching package: 'caret'

## The following object is masked from 'package:purrr':
##
##     lift
```

##Question1.)Use only the numerical variables (1 to 9) to cluster the 21 firms. Justify the various choices made in conducting the cluster analysis, such as weights for different variables, the specific clustering algorithm(s) used, the number of clusters formed, and so on.

#Implementing z-score scaling to the pharmaceuticals dataframe.

```r
Scaled_input_pharmaceuticals<- scale(input_pharmaceuticals[,3:11])
summary(Scaled_input_pharmaceuticals)
```

```
##    Market_Cap           Beta            PE_Ratio            ROE
##  Min.   :-0.9768   Min.   :-1.3466   Min.   :-1.3404   Min.   :-1.4515
##  1st Qu.:-0.8763   1st Qu.:-0.6844   1st Qu.:-0.4023   1st Qu.:-0.7223
##  Median :-0.1614   Median :-0.2560   Median :-0.2429   Median :-0.2118
##  Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000
##  3rd Qu.: 0.2762   3rd Qu.: 0.4841   3rd Qu.: 0.1495   3rd Qu.: 0.3450
##  Max.   : 2.4200   Max.   : 2.2758   Max.   : 3.4971   Max.   : 2.4597
##       ROA           Asset_Turnover       Leverage          Rev_Growth
##  Min.   :-1.7128   Min.   :-1.8451   Min.   :-0.74966   Min.   :-1.4971
##  1st Qu.:-0.9047   1st Qu.:-0.4613   1st Qu.:-0.54487   1st Qu.:-0.6328
##  Median : 0.1289   Median :-0.4613   Median :-0.31449   Median :-0.3621
##  Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.00000   Mean   : 0.0000
##  3rd Qu.: 0.8430   3rd Qu.: 0.9225   3rd Qu.: 0.01828   3rd Qu.: 0.7693
##  Max.   : 1.8389   Max.   : 1.8451   Max.   : 3.74280   Max.   : 1.8862
##  Net_Profit_Margin
##  Min.   :-1.99560
##  1st Qu.:-0.68504
##  Median : 0.06168
##  Mean   : 0.00000
##  3rd Qu.: 0.82364
##  Max.   : 1.49416
```

#Range scaling the dataframe is being done.

```r
range_input_pharmaceuticals <- scale(input_pharmaceuticals[,3:11])
```

#the power source scaled data frame's summary

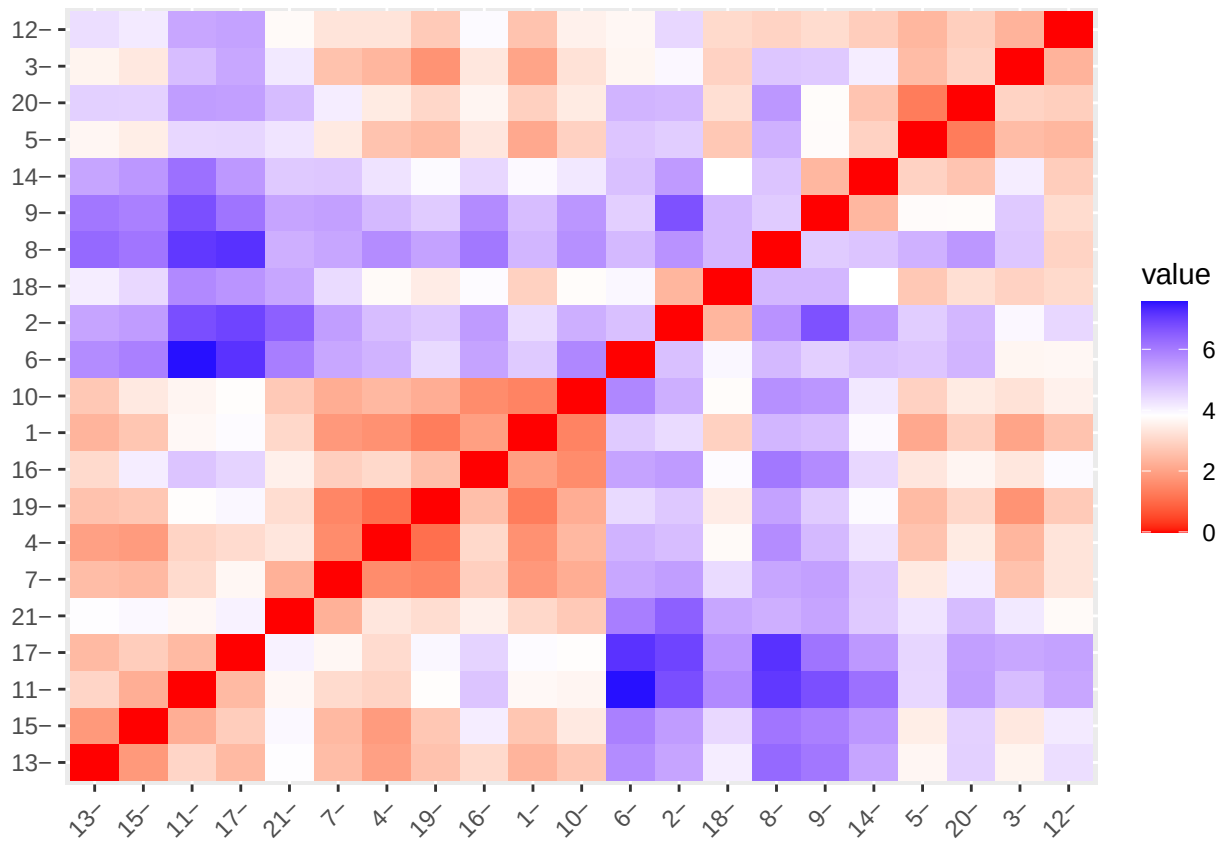
```r
summary(range_input_pharmaceuticals)
```

```
##    Market_Cap           Beta            PE_Ratio            ROE
##  Min.   :-0.9768   Min.   :-1.3466   Min.   :-1.3404   Min.   :-1.4515
##  1st Qu.:-0.8763   1st Qu.:-0.6844   1st Qu.:-0.4023   1st Qu.:-0.7223
##  Median :-0.1614   Median :-0.2560   Median :-0.2429   Median :-0.2118
##  Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000
##  3rd Qu.: 0.2762   3rd Qu.: 0.4841   3rd Qu.: 0.1495   3rd Qu.: 0.3450
##  Max.   : 2.4200   Max.   : 2.2758   Max.   : 3.4971   Max.   : 2.4597
##       ROA           Asset_Turnover       Leverage          Rev_Growth
##  Min.   :-1.7128   Min.   :-1.8451   Min.   :-0.74966   Min.   :-1.4971
##  1st Qu.:-0.9047   1st Qu.:-0.4613   1st Qu.:-0.54487   1st Qu.:-0.6328
##  Median : 0.1289   Median :-0.4613   Median :-0.31449   Median :-0.3621
##  Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.00000   Mean   : 0.0000
##  3rd Qu.: 0.8430   3rd Qu.: 0.9225   3rd Qu.: 0.01828   3rd Qu.: 0.7693
##  Max.   : 1.8389   Max.   : 1.8451   Max.   : 3.74280   Max.   : 1.8862
##  Net_Profit_Margin
##  Min.   :-1.99560
##  1st Qu.:-0.68504
##  Median : 0.06168
##  Mean   : 0.00000
```

```
##  3rd Qu.: 0.82364
##  Max.   : 1.49416
```

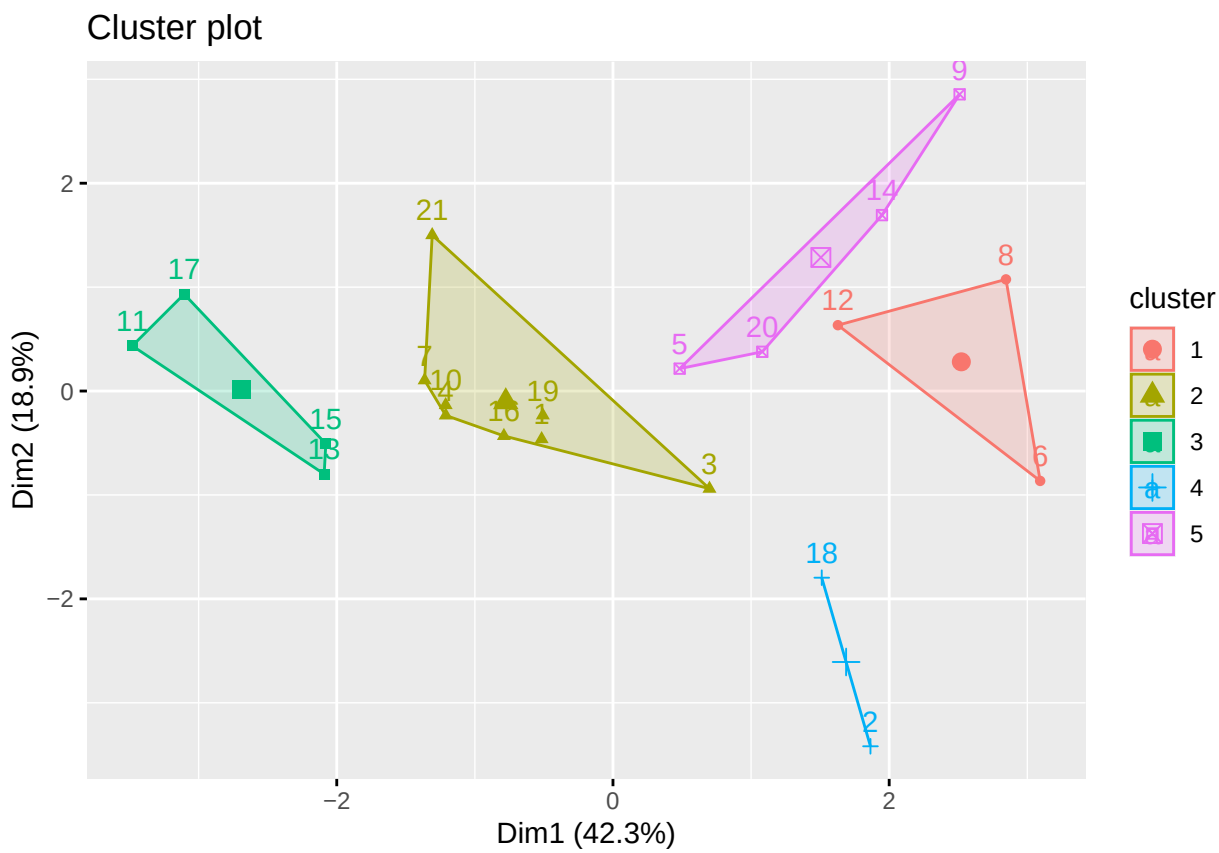
#the scaled medicines data's measurement of distance.

```
distancy <- get_dist(Scaled_input_pharmaceuticals)

fviz_dist(distancy) #illustrating the separation between the distance matrix's rows
```



#scaling the data and using K-means clustering to organize it.

```
kmeans_R1 <- kmeans(Scaled_input_pharmaceuticals, centers = 5, nstart = 25)

#The graph visualization of the clusters
fviz_cluster(kmeans_R1, data = Scaled_input_pharmaceuticals)
```




```r
print(kmeans_R1)
```

```
## K-means clustering with 5 clusters of sizes 3, 8, 4, 2, 4
##
## Cluster means:
##    Market_Cap       Beta    PE_Ratio        ROE        ROA Asset_Turnover
## 1 -0.87051511  1.3409869 -0.05284434 -0.6184015 -1.1928478     -0.4612656
## 2 -0.03142211 -0.4360989 -0.31724852  0.1950459  0.4083915      0.1729746
## 3  1.69558112 -0.1780563 -0.19845823  1.2349879  1.3503431      1.1531640
## 4 -0.43925134 -0.4701800  2.70002464 -0.8349525 -0.9234951      0.2306328
## 5 -0.76022489  0.2796041 -0.47742380 -0.7438022 -0.8107428     -1.2684804
##      Leverage Rev_Growth Net_Profit_Margin
## 1  1.36644699 -0.6912914      -1.320000179
## 2 -0.27449312 -0.7041516       0.556954446
## 3 -0.46807818  0.4671788       0.591242521
## 4 -0.14170336 -0.1168459      -1.416514761
## 5  0.06308085  1.5180158      -0.006893899
##
## Clustering vector:
##  [1] 2 4 2 2 5 1 2 1 5 2 3 1 3 5 3 2 3 4 2 5 2
##
## Within cluster sum of squares by cluster:
## [1] 15.595925 21.879320  9.284424  2.803505 12.791257
##  (between_SS / total_SS =  65.4 %)
##
```

```
## Available components:
##
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
```

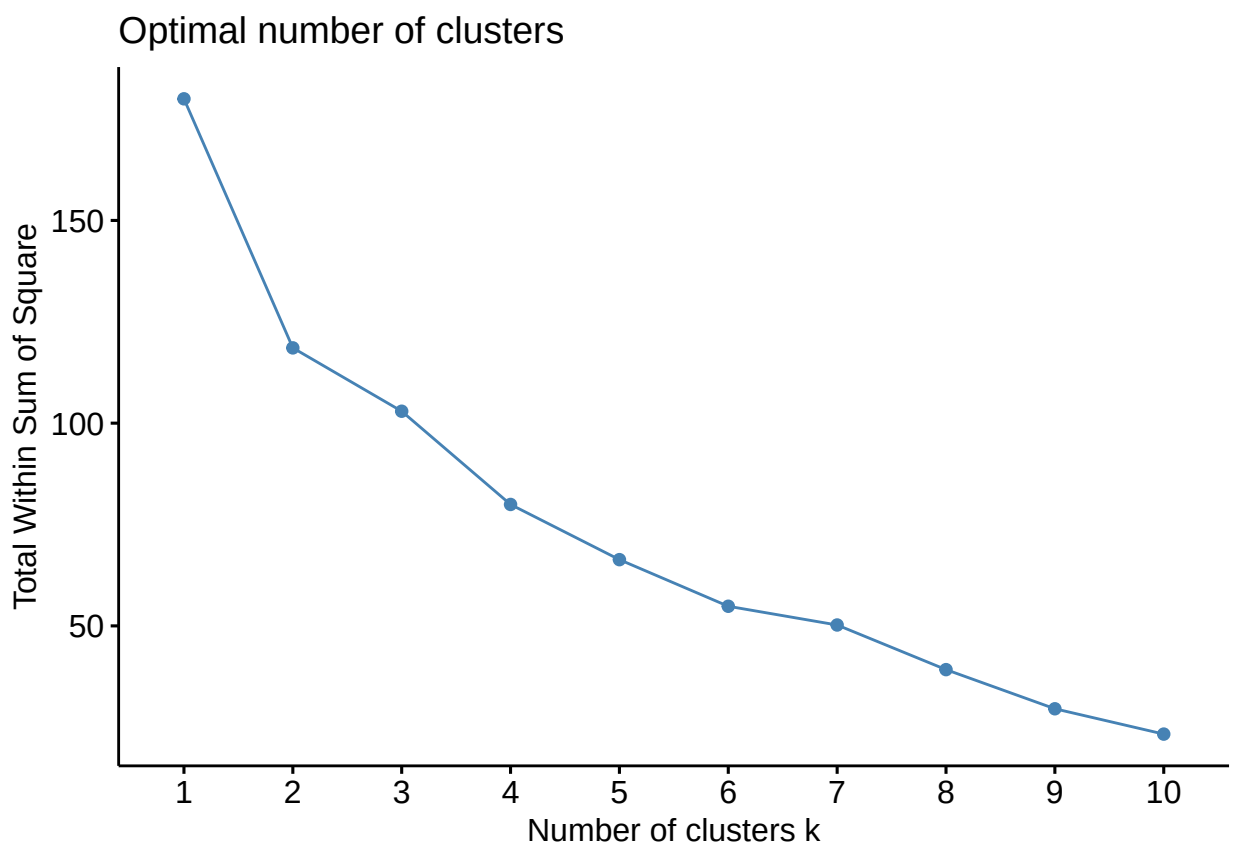
##OVERVIEW: K-means Clustering

##OverviewIn -Input on nine factors associated with each among the twenty-one pharmaceutical businesses can be obtained via the summary algorithm applied to the k-means data. There are five different clusters in all, each with five centroid points, and the numbers are 2, 3, 4, 4, 8, and 9.

##Question2.a) Interpret the clusters with respect to the numerical variables used in forming the clusters.

#charting the sum of the values vs the total number of clusters

```
fviz_nbclust(Scaled_input_pharmaceuticals, kmeans, method = "wss")
```

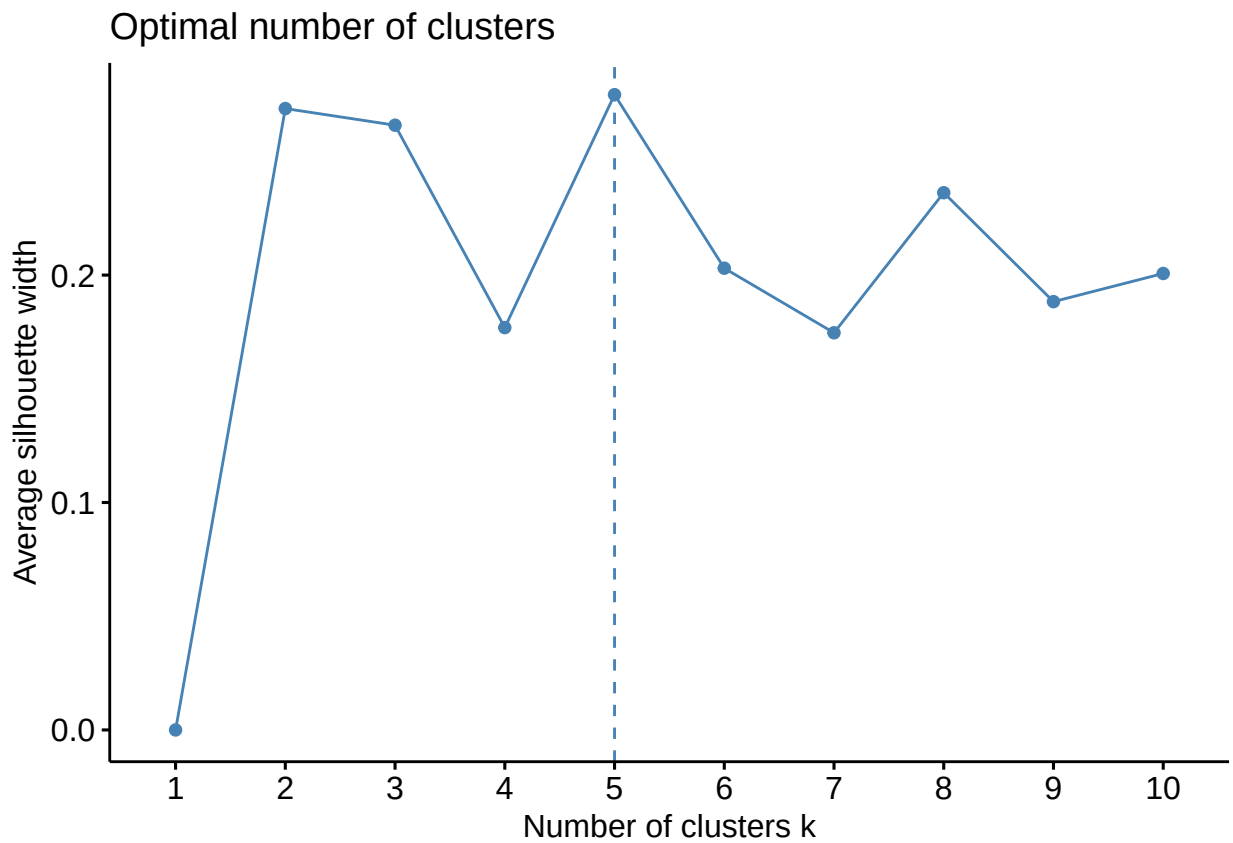


In this plot, it's evident that the graph is not forming an elbow shape and is slightly flattening at k=4 and k=6.

#Plotting the amount of clusters will allow you to determine how many are optimum.

#compared to the typical silence width.

```
fviz_nbclust(Scaled_input_pharmaceuticals, kmeans, method = "silhouette")
```



##We can quickly determine that the most effective and most optimum option for this k-means clustering is 5 clusters based on the Silhoutte graph.

##Question 2.b) Is there a pattern in the clusters with respect to the numerical variables (10 to 12)? (those not used in forming the clusters).

#determining the mean value based on the data that is shown in the clusters.

#carrying out the pharmaceutical data's aggregate functionality.

```
aggregate(input_pharmaceuticals[3:11], by=list(cluster=kmeans_R1$cluster), mean)
```

```
##   cluster Market_Cap    Beta PE_Ratio      ROE       ROA Asset_Turnover
## 1       1   6.636667 0.87000  24.6000 16.46667  4.166667         0.6000
## 2       2  55.810000 0.41375  20.2875 28.73750 12.687500         0.7375
## 3       3 157.017500 0.48000  22.2250 44.42500 17.700000         0.9500
## 4       4  31.910000 0.40500  69.5000 13.20000  5.600000         0.7500
## 5       5  13.100000 0.59750  17.6750 14.57500  6.200000         0.4250
##   Leverage Rev_Growth Net_Profit_Margin
## 1 1.653333   5.733333          7.033333
## 2 0.371250   5.591250         19.350000
## 3 0.220000  18.532500         19.575000
## 4 0.475000  12.080000          6.400000
## 5 0.635000  30.142500         15.650000
```

#employ Cbind to do the data frame merging.

```r
Short_term_data <- cbind(input_pharmaceuticals, cluster = kmeans_R1$cluster)
tibble(Short_term_data)
```

```
## # A tibble: 21 x 15
##    Symbol Name     Market_Cap  Beta PE_Ratio   ROE   ROA Asset_Turnover Leverage
##    <chr>  <chr>         <dbl> <dbl>    <dbl> <dbl> <dbl>          <dbl>    <dbl>
##  1 ABT    Abbott ~      68.4   0.32     24.7  26.4  11.8            0.7     0.42
##  2 AGN    Allerga~       7.58  0.41     82.5  12.9   5.5            0.9     0.6
##  3 AHM    Amersha~       6.3   0.46     20.7  14.9   7.8            0.9     0.27
##  4 AZN    AstraZe~      67.6   0.52     21.5  27.4  15.4            0.9     0
##  5 AVE    Aventis       47.2   0.32     20.1  21.8   7.5            0.6     0.34
##  6 BAY    Bayer AG      16.9   1.11     27.9   3.9   1.4            0.6     0
##  7 BMY    Bristol~      51.3   0.5      13.9  34.8  15.1            0.9     0.57
##  8 CHTT   Chattem~       0.41  0.85     26    24.1   4.3            0.6     3.51
##  9 ELN    Elan Co~       0.78  1.08      3.6  15.1   5.1            0.3     1.07
## 10 LLY    Eli Lil~      73.8   0.18     27.9  31    13.5            0.6     0.53
## # i 11 more rows
## # i 6 more variables: Rev_Growth <dbl>, Net_Profit_Margin <dbl>,
## #   Median_Recommendation <chr>, Location <chr>, Exchange <chr>, cluster <int>
```

#Summarizing the particular beakdown by cluster

```r
by(Short_term_data, factor(Short_term_data$cluster), summary)
```

```
## factor(Short_term_data$cluster): 1
##     Symbol              Name             Market_Cap          Beta
##  Length:3           Length:3           Min.   : 0.410   Min.   :0.65
##  Class :character   Class :character   1st Qu.: 1.505   1st Qu.:0.75
##  Mode  :character   Mode  :character   Median : 2.600   Median :0.85
##                                        Mean   : 6.637   Mean   :0.87
##                                        3rd Qu.: 9.750   3rd Qu.:0.98
##                                        Max.   :16.900   Max.   :1.11
##     PE_Ratio          ROE             ROA        Asset_Turnover    Leverage
##  Min.   :19.90   Min.   : 3.90   Min.   :1.400   Min.   :0.6    Min.   :0.000
##  1st Qu.:22.95   1st Qu.:12.65   1st Qu.:2.850   1st Qu.:0.6    1st Qu.:0.725
##  Median :26.00   Median :21.40   Median :4.300   Median :0.6    Median :1.450
##  Mean   :24.60   Mean   :16.47   Mean   :4.167   Mean   :0.6    Mean   :1.653
##  3rd Qu.:26.95   3rd Qu.:22.75   3rd Qu.:5.550   3rd Qu.:0.6    3rd Qu.:2.480
##  Max.   :27.90   Max.   :24.10   Max.   :6.800   Max.   :0.6    Max.   :3.510
##    Rev_Growth     Net_Profit_Margin Median_Recommendation   Location
##  Min.   :-3.170   Min.   : 2.600    Length:3              Length:3
##  1st Qu.: 1.605   1st Qu.: 5.050    Class :character      Class :character
##  Median : 6.380   Median : 7.500    Mode  :character      Mode  :character
##  Mean   : 5.733   Mean   : 7.033
##  3rd Qu.:10.185   3rd Qu.: 9.250
##  Max.   :13.990   Max.   :11.000
##    Exchange            cluster
##  Length:3           Min.   :1
##  Class :character   1st Qu.:1
##  Mode  :character   Median :1
##                     Mean   :1
##                     3rd Qu.:1
```

```
##                     Max.   :1
## ------------------------------------------------------------
## factor(Short_term_data$cluster): 2
##     Symbol              Name             Market_Cap         Beta
##  Length:8           Length:8           Min.   : 6.30   Min.   :0.1800
##  Class :character   Class :character   1st Qu.:44.67   1st Qu.:0.2875
##  Mode  :character   Mode  :character   Median :59.48   Median :0.4800
##                                        Mean   :55.81   Mean   :0.4138
##                                        3rd Qu.:69.79   3rd Qu.:0.5125
##                                        Max.   :96.65   Max.   :0.6300
##     PE_Ratio          ROE             ROA        Asset_Turnover
##  Min.   :13.10   Min.   :14.90   Min.   : 7.80   Min.   :0.5000
##  1st Qu.:17.65   1st Qu.:21.43   1st Qu.:11.65   1st Qu.:0.6000
##  Median :21.10   Median :26.90   Median :13.35   Median :0.7500
##  Mean   :20.29   Mean   :28.74   Mean   :12.69   Mean   :0.7375
##  3rd Qu.:22.38   3rd Qu.:31.95   3rd Qu.:13.90   3rd Qu.:0.9000
##  Max.   :27.90   Max.   :54.90   Max.   :15.40   Max.   :0.9000
##     Leverage        Rev_Growth     Net_Profit_Margin Median_Recommendation
##  Min.   :0.0000   Min.   :-2.690   Min.   :11.20     Length:8
##  1st Qu.:0.0450   1st Qu.: 2.115   1st Qu.:17.23     Class :character
##  Median :0.3450   Median : 6.630   Median :19.30     Mode  :character
##  Mean   :0.3713   Mean   : 5.591   Mean   :19.35
##  3rd Qu.:0.5400   3rd Qu.: 7.795   3rd Qu.:22.65
##  Max.   :1.1200   Max.   :15.000   Max.   :25.50
##    Location           Exchange            cluster
##  Length:8           Length:8           Min.   :2
##  Class :character   Class :character   1st Qu.:2
##  Mode  :character   Mode  :character   Median :2
##                                        Mean   :2
##                                        3rd Qu.:2
##                                        Max.   :2
## ------------------------------------------------------------
## factor(Short_term_data$cluster): 3
##     Symbol              Name             Market_Cap         Beta
##  Length:4           Length:4           Min.   :122.1   Min.   :0.3500
##  Class :character   Class :character   1st Qu.:129.9   1st Qu.:0.4325
##  Mode  :character   Mode  :character   Median :153.2   Median :0.4600
##                                        Mean   :157.0   Mean   :0.4800
##                                        3rd Qu.:180.3   3rd Qu.:0.5075
##                                        Max.   :199.5   Max.   :0.6500
##     PE_Ratio          ROE             ROA        Asset_Turnover
##  Min.   :18.00   Min.   :28.60   Min.   :15.00   Min.   :0.800
##  1st Qu.:18.68   1st Qu.:37.60   1st Qu.:15.97   1st Qu.:0.875
##  Median :21.25   Median :43.10   Median :17.75   Median :0.950
##  Mean   :22.23   Mean   :44.42   Mean   :17.70   Mean   :0.950
##  3rd Qu.:24.80   3rd Qu.:49.92   3rd Qu.:19.48   3rd Qu.:1.025
##  Max.   :28.40   Max.   :62.90   Max.   :20.30   Max.   :1.100
##     Leverage       Rev_Growth    Net_Profit_Margin Median_Recommendation
##  Min.   :0.100   Min.   : 9.37   Min.   :14.10     Length:4
##  1st Qu.:0.145   1st Qu.:15.36   1st Qu.:16.95     Class :character
##  Median :0.220   Median :19.61   Median :19.50     Mode  :character
##  Mean   :0.220   Mean   :18.53   Mean   :19.57
##  3rd Qu.:0.295   3rd Qu.:22.79   3rd Qu.:22.12
##  Max.   :0.340   Max.   :25.54   Max.   :25.20
```

```
##    Location           Exchange            cluster
##  Length:4           Length:4           Min.   :3
##  Class :character   Class :character   1st Qu.:3
##  Mode  :character   Mode  :character   Median :3
##                                        Mean   :3
##                                        3rd Qu.:3
##                                        Max.   :3
## ------------------------------------------------------------
## factor(Short_term_data$cluster): 4
##     Symbol              Name             Market_Cap         Beta
##  Length:2           Length:2           Min.   : 7.58   Min.   :0.4000
##  Class :character   Class :character   1st Qu.:19.75   1st Qu.:0.4025
##  Mode  :character   Mode  :character   Median :31.91   Median :0.4050
##                                        Mean   :31.91   Mean   :0.4050
##                                        3rd Qu.:44.08   3rd Qu.:0.4075
##                                        Max.   :56.24   Max.   :0.4100
##     PE_Ratio         ROE             ROA       Asset_Turnover     Leverage
##  Min.   :56.5   Min.   :12.90   Min.   :5.50   Min.   :0.600   Min.   :0.3500
##  1st Qu.:63.0   1st Qu.:13.05   1st Qu.:5.55   1st Qu.:0.675   1st Qu.:0.4125
##  Median :69.5   Median :13.20   Median :5.60   Median :0.750   Median :0.4750
##  Mean   :69.5   Mean   :13.20   Mean   :5.60   Mean   :0.750   Mean   :0.4750
##  3rd Qu.:76.0   3rd Qu.:13.35   3rd Qu.:5.65   3rd Qu.:0.825   3rd Qu.:0.5375
##  Max.   :82.5   Max.   :13.50   Max.   :5.70   Max.   :0.900   Max.   :0.6000
##    Rev_Growth    Net_Profit_Margin Median_Recommendation   Location
##  Min.   : 9.16   Min.   :5.50      Length:2              Length:2
##  1st Qu.:10.62   1st Qu.:5.95      Class :character      Class :character
##  Median :12.08   Median :6.40      Mode  :character      Mode  :character
##  Mean   :12.08   Mean   :6.40
##  3rd Qu.:13.54   3rd Qu.:6.85
##  Max.   :15.00   Max.   :7.30
##    Exchange            cluster
##  Length:2           Min.   :4
##  Class :character   1st Qu.:4
##  Mode  :character   Median :4
##                     Mean   :4
##                     3rd Qu.:4
##                     Max.   :4
## ------------------------------------------------------------
## factor(Short_term_data$cluster): 5
##     Symbol              Name             Market_Cap          Beta
##  Length:4           Length:4           Min.   : 0.780   Min.   :0.2400
##  Class :character   Class :character   1st Qu.: 1.095   1st Qu.:0.3000
##  Mode  :character   Mode  :character   Median : 2.230   Median :0.5350
##                                        Mean   :13.100   Mean   :0.5975
##                                        3rd Qu.:14.235   3rd Qu.:0.8325
##                                        Max.   :47.160   Max.   :1.0800
##     PE_Ratio          ROE             ROA        Asset_Turnover
##  Min.   : 3.60   Min.   :10.20   Min.   :5.100   Min.   :0.300
##  1st Qu.:14.70   1st Qu.:10.95   1st Qu.:5.325   1st Qu.:0.300
##  Median :19.25   Median :13.15   Median :6.100   Median :0.400
##  Mean   :17.68   Mean   :14.57   Mean   :6.200   Mean   :0.425
##  3rd Qu.:22.23   3rd Qu.:16.77   3rd Qu.:6.975   3rd Qu.:0.525
##  Max.   :28.60   Max.   :21.80   Max.   :7.500   Max.   :0.600
##     Leverage       Rev_Growth    Net_Profit_Margin Median_Recommendation
```

```
##  Min.   :0.200   Min.   :26.81   Min.   :12.90     Length:4
##  1st Qu.:0.305   1st Qu.:28.59   1st Qu.:13.20     Class :character
##  Median :0.635   Median :29.77   Median :14.20     Mode  :character
##  Mean   :0.635   Mean   :30.14   Mean   :15.65
##  3rd Qu.:0.965   3rd Qu.:31.33   3rd Qu.:16.65
##  Max.   :1.070   Max.   :34.21   Max.   :21.30
##    Location           Exchange            cluster
##  Length:4           Length:4           Min.   :5
##  Class :character   Class :character   1st Qu.:5
##  Mode  :character   Mode  :character   Median :5
##                                        Mean   :5
##                                        3rd Qu.:5
##                                        Max.   :5
```

#median the calculation

```
Commend_tables <- table(Short_term_data$cluster, Short_term_data$Median_Recommendation)
names(dimnames(Commend_tables)) <- c("Cluster", "Recommendation")
Commend_tables <- addmargins(Commend_tables)
Commend_tables
```

```
##        Recommendation
## Cluster Hold Moderate Buy Moderate Sell Strong Buy Sum
##     1      2            1             0          0   3
##     2      4            1             2          1   8
##     3      2            2             0          0   4
##     4      1            1             0          0   2
##     5      0            2             2          0   4
##     Sum    9            7             4          1  21
```

##There are 21 suggestions in total: one strong buy, seven moderate buys, nine holds, and four moderate sells. Cluster 5 combines all four suggestions, including the opposing advise on buys and sells. Clusters 1, 2, and 3 only include information about mod purchases and holds.For Cluster 4, there is a moderate buy and a moderate sell suggestion.

#Cluster breakdown of the organization's headquarters based on the combined data.

```
Loctn_Table <- table(Short_term_data$cluster, Short_term_data$Location)
names(dimnames(Loctn_Table)) <- c("Cluster", "Location")
Loctn_Table <- addmargins(Loctn_Table)
Loctn_Table
```

```
##        Location
## Cluster CANADA FRANCE GERMANY IRELAND SWITZERLAND UK US Sum
##     1        0      0       1       0           0  0  2   3
##     2        0      0       0       0           1  2  5   8
##     3        0      0       0       0           0  1  3   4
##     4        1      0       0       0           0  0  1   2
##     5        0      1       0       1           0  0  2   4
##     Sum      1      1       1       1           1  3 13  21
```

##There are a total of 21 firms: 13 are located in the United States, 3 in the United Kingdom, and 1 in each of Canada, France, Germany, Ireland, and Switzerland. Cluster 5 features the US, UK, and Switzerland.

Cluster 2 has the US and Canada. Cluster 3 features the US and Britain. Cluster 4 features the US, France, and Ireland.

#pulling up a summary each of the cluster's stock exchange values.

#establishing the exchange table and generating a data frame for the combined information.

```
Exg_Tables <- table(Short_term_data$cluster, Short_term_data$Exchange)
names(dimnames(Exg_Tables)) <- c("Cluster", "Exchange")
Exg_Tables <- addmargins(Exg_Tables)
Exg_Tables
```

```
##        Exchange
## Cluster AMEX NASDAQ NYSE Sum
##     1      1      1    1   3
##     2      0      0    8   8
##     3      0      0    4   4
##     4      0      0    2   2
##     5      0      0    4   4
##     Sum    1      1   19  21
```

##In total, there are 21 companies: 1 Amex, 1 Nasdaq, and 19 NYSE. Just the NYSE is in Cluster 5. Cluster 2 includes all three. The only thing in clusters 1, 3, and 4 is NYSE.

##Question 3.) Provide an appropriate name for each cluster using any or all of the variables in the dataset.

Answers :- Cluster 1: - All US firms listed on the NYSE that are either held or purchased make up Cluster 1: "High Market Cap - High RoE" greater NetProfitMargin.

Cluster 2: This cluster, which includes exclusively NYSE-listed businesses, is dubbed "Hold CLuster"

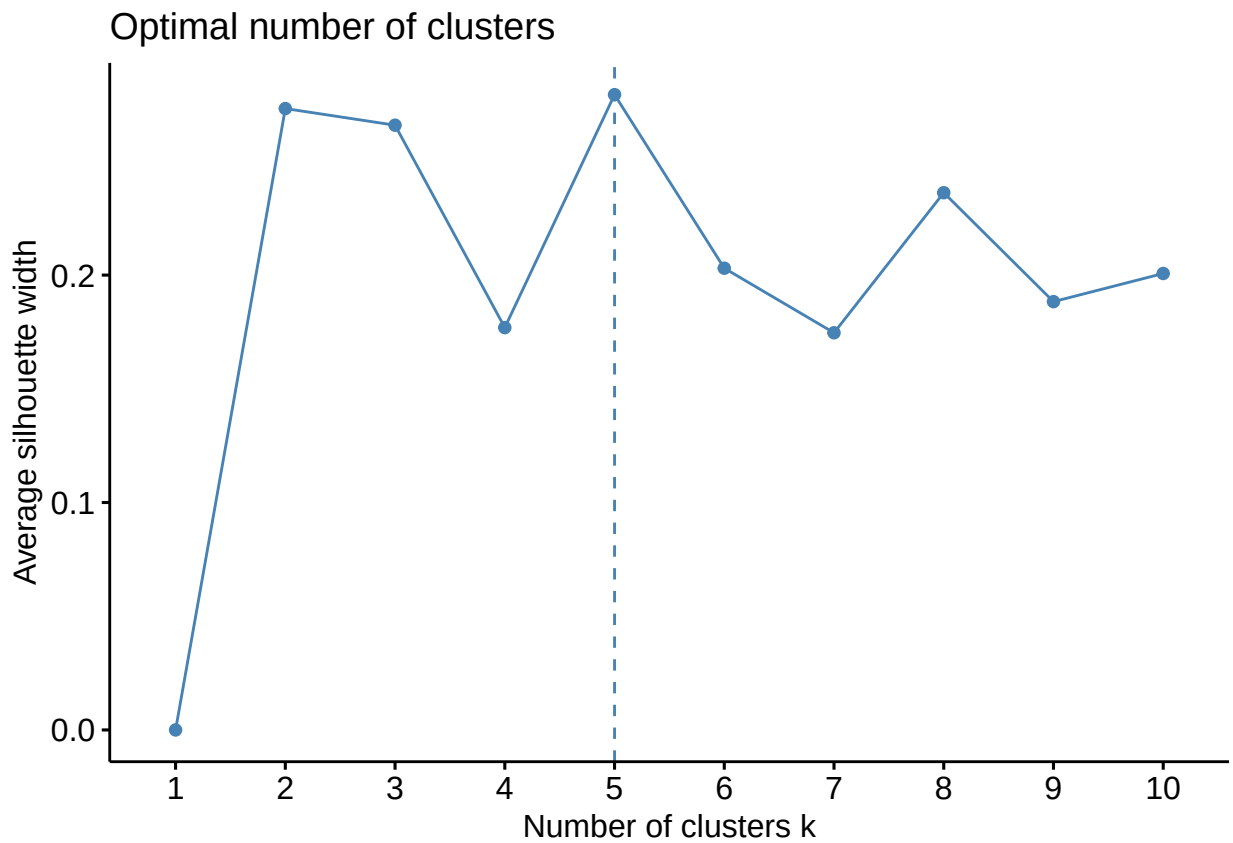
Cluster 3: This is called "Low Market Cap & Less ROA"; on stock exchanges, it is either buy or hold.

Cluster 4:This fits under the heading of "High PE ratio / Small Net Profit Margin.".

Cluster 5: The moniker "Moderate Buy & Moderate Sell" might be used to this cluster.

##Trying or investigating alternative techniques to determine if they may provide more effective clustering as well?

```
fviz_nbclust(range_input_pharmaceuticals, FUN = kmeans, method = "silhouette")
```




```
kmeans_R2 = kcca(Scaled_input_pharmaceuticals, k=5, kccaFamily("kmeans"))
kmeans_R2
```

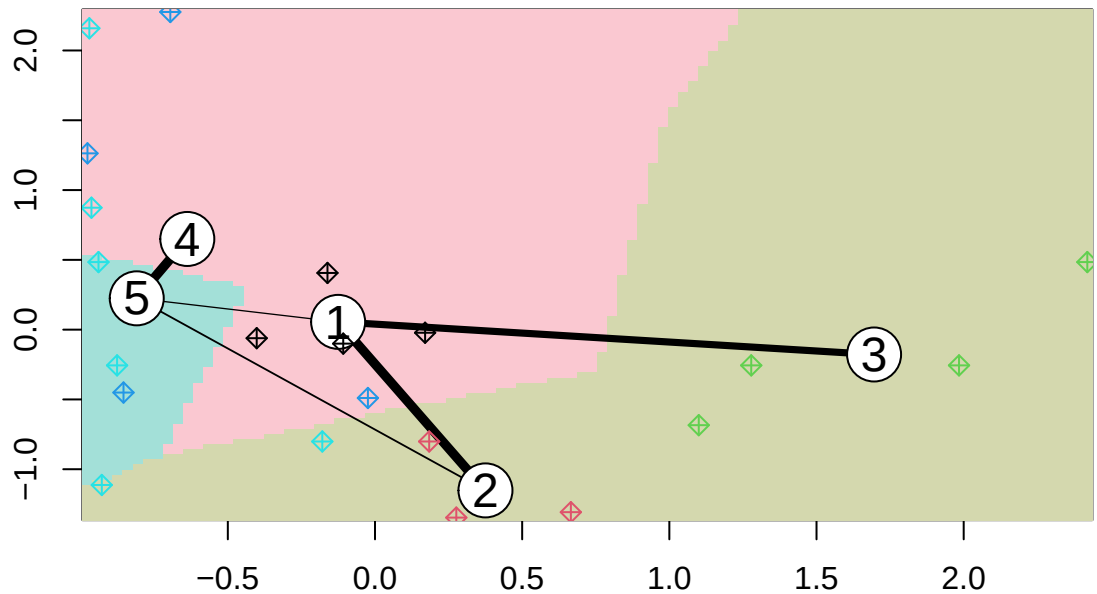
```
## kcca object of family 'kmeans'
##
## call:
## kcca(x = Scaled_input_pharmaceuticals, k = 5, family = kccaFamily("kmeans"))
##
## cluster sizes:
##
## 1 2 3 4 5
## 4 3 4 4 6
```

```
clusters(kmeans_R2)
```

```
##  [1] 2 4 5 1 5 4 1 4 5 2 3 5 3 5 3 2 3 4 1 5 1
```

#Using the technique of predict().

```
index_clusters <- predict(kmeans_R2)
image(kmeans_R2)
points(Scaled_input_pharmaceuticals, col=index_clusters, pch=9, cex=1.0)
```



##In this instance, the kcca method is utilized to perform a kmeans cluster on k = 5, as opposed to the kmeans function from basic R. In comparison to the base R method, the clustering has the same size but a distinct assignment between the points. The clustering graph shows that there is more ambiguity in the grouping than we would want, especially between groups 1, 2, and 3.

PLOTTING THE CLUSTERS AND THE KMEDIANS.

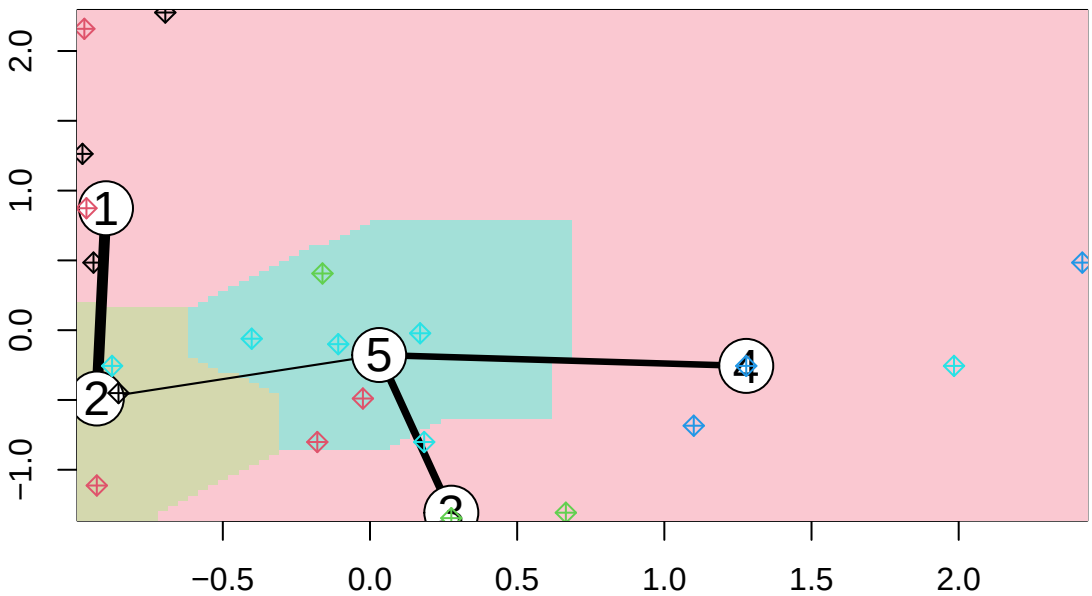
```
kmeans_R2 = kcca(Scaled_input_pharmaceuticals, k=5, kccaFamily("kmedians"))
kmeans_R2
```

```
## kcca object of family 'kmedians'
##
## call:
## kcca(x = Scaled_input_pharmaceuticals, k = 5, family = kccaFamily("kmedians"))
##
## cluster sizes:
##
## 1 2 3 4 5
## 4 5 3 3 6
```

```
clusters(kmeans_R2) #clusteing
```

```
##  [1] 5 1 5 5 2 1 5 1 2 3 4 1 5 2 4 3 4 2 5 2 3
```

```r
index_clusters <- predict(kmeans_R2)
image(kmeans_R2)
points(Scaled_input_pharmaceuticals, col=index_clusters, pch=9, cex=1.0)
```



##The sizes of the five clusters are 4, 5, 3, 3, and 6 if we convert the KCC to Kmedian from Kmeans. However, the clustering is less pronounced. Though it's uncertain if a better cluster exists, we plan to look into the additional data to see if there are any other methods or resources we might employ to improve the visual cluster.